A Deformable Attention Network for High-Resolution Remote Sensing Images Semantic Segmentation

Renxiang Zuo, Guangyun Zhang, Rongting Zhang, and Xiuping Jia, *Fellow, IEEE*

***Abstract*— Deformable convolutional networks (DCNs) can mitigate the inherent limited geometric transformation. We reformulate the spatialwise attention mechanism using DCNs in this article for semantic segmentation of high-resolution remote sensing (HRRS) images. It combines the sparse spatial sampling strategy and the long-range relationship modeling capability, namely, deformable attention module (DAM). Such locality awareness, more adaptable to HRRS image structures, can capture each pixel's neighboring structural information. A reasonable multiscale deformable attention net (MDANet) is designed for the HRRS image semantic segmentation with a slightly increased computational cost based on the proposed DAM. Specifically, standard convolutional layers in the raw ResNet50 are equipped with a DAM to control sampling over a broader range of feature levels and aggregate multiscale context information. The experimental results evaluated on Vaihingen and DeepGlobe Land Cover Classification datasets show that the performance accuracy of MDANet is improved by 7.77% and 8.45% compared with the backbone network (ResNet50) in terms of Miou evaluation, respectively. Furthermore, a DAM can perform better than a global spatial attention mechanism with less computation on the 3 × 64 × 64 feature map. In addition, the added ablation studies demonstrate the effectiveness and efficiency of the DAM and multiscale strategy, respectively. Moreover, the sensitivity of critical hyperparameters is analyzed.**

***Index Terms*— Deformable convolutional networks (DCNs), high-resolution remote sensing (HRRS) images, multiscale, spatialwise attention mechanism.**

I. Introduction

With the rapid development of satellite and airborne remote sensing technology over recent years [1], high-resolution remote sensing (HRRS) images can be efficiently acquired and are widely used for archeological and cultural heritage [2], earth environmental monitoring [3], coastal management [4], urban forestry [5], agricultural practices [6], image retrieval [7], and so on. Compared with low- and medium-resolution data, HRRS images obtain massive fine details. On the other hand, they often present redundant and noisy ground object details due to the imaging characteristics.

In addition, many multiple ground objects in HRRS images are challenging to be distinguished due to the high intraclass variance (e.g., low vegetation and tree) and low interclass (e.g., building) separability. Moreover, shadows of multiple ground objects and the multiscale phenomenon raise more diversity in spatial disturbances. As a result, it is challenging to describe and extract effective spatial information from HRRS images.

During the last decade, extensive visual recognition algorithms [8] that are relied on human-engineering descriptors were developed, such as color histogram (CH), histogram of oriented gradients (HOG), scale-invariant feature transformation (SIFT), and so on. Moreover, unsupervised learning methods can automatically learn features from unlabeled images, such as principal component analysis (PCA), k-means clustering, and sparse coding. However, the representation capability is limited by handcrafted features and dependence on the high professional knowledge of the domain experts. The potential of HRRS images' characteristics cannot be fully unleashed. Due to sufficient parallel computing resources and the booming of deep learning (DL) theory [9], especially deep convolution neural networks (DCNNs), hierarchical contextual visual features could be automagical learned instead of defining in advance. DCNN-based algorithms [10] have increasingly prevailed in the remote sensing field during the past few years.

Generally, the DCNN-based methods developed to HRRS image semantic segmentation can be summarized into three categories: 1) pretrained DCNNs are directly applied as feature extractors with parameters trained on other large datasets [11]; 2) pretrained DCNNs are fine-tuned on HRRS datasets [12]; and 3) the HRRS datasets are fed to train a DCNN from scratch [13]. Despite DCNNs' powerful feature extraction capability, they are inherently unable to model long-range dependence or simulating geometric transformation. Learning to capture long-range dependence in data is a fundamental problem in DL. Pioneering works [14]–[17] mainly focus on enlarging the receptive fields by applying large kernel sizes, dilated convolutions, feature pyramids, or stacking multiple convolutions. They have some limitations. For example, stacking multiple convolution operators may cause inefficient

Manuscript received September 6, 2021; accepted October 4, 2021. Date of publication October 13, 2021; date of current version February 7, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2018YFC1407400 and in part by the National Natural Science Foundation of China under Grant 41601365. *(Corresponding author: Guangyun Zhang.)*

Renxiang Zuo, Guangyun Zhang, and Rongting Zhang are with the School of Geomatics Science and Technology, Nanjing Tech University, Nanjing 211816, China (e-mail: renxiang@njtech.edu.cn; zgy@njtech.edu.cn; zrt@njtech.edu.cn).

Xiuping Jia is with the School of Engineering and Information Technology, University of New South Wales, Canberra, ACT 2612, Australia (e-mail: x.jia@adfa.edu.au).

Digital Object Identifier 10.1109/TGRS.2021.3119537

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

reasoning, higher computational load, and optimization difficulties during the training.

Given the overwhelming computational requirement for processing the whole scene, the human visual system (HVS) endows humans to selectively attend to the information most relevant to the behavior of a visual stimulus. A simple understanding is that the HVS allocates computing resources to the most valuable and informative feature areas. For example, when quickly scanning a whole HRRS image, humans automatically observe the overall spatial structures, focusing on the most effective information of the task-relevant areas and suppressing other redundant information. With the guidance of the HVS, recent DL models [18] encode HVS to allow the network to flexibly control limited computing resources to the most task-relevant information through pairwise interactions regardless of the spatial distance. Mathematically, the self-attention mechanism learns weights via pairwise dot products over the input signal and then sums all signals up with corresponding weights. There are, however, two limitations in the self-attention mechanism.

1) *Memory Bottleneck:* The principle of the self-attention mechanism is to obtain the similarity matrix between any two pixels. However, the semantic segmentation task requires dense prediction at the pixel level. An unacceptable quadratic memory and computational complexity inhibit directly modeling global relations on HRRS images.
2) *Local Insensitivity:* For HRRS images, it is natural that neighboring spatial pixels are usually highly correlated, and most dependence relate to nearby neighborhoods of pixels. However, the self-attention is insensitive to the local context and may disperse attention weights due to the weighted sum operations. The phenomenon results in the range of interest exceed the boundary of objects and bring noisy attention weights. Hence, effectively combining global and local information may be more efficient.

Though visual recognition has developed rapidly with DCNNs, feature extraction still suffers from the relatively common multiscale phenomenon. The multiscale phenomenon generalizes that scales of objects and features from multiple layers vary significantly. Naturally, different objects are peculiarly presented at various scales and irregular shapes between different categories and between the same categories. Besides, redundant object details (e.g., building shadow and tree branches) lead to complex boundaries, as shown in Fig. 1. Consequently, the multiscale phenomenon of complex objects with large-scale variations may cause challenges in distinguishing various target objects relying on a single-sized fixed receptive field only. Specifically, the too-large receptive field will introduce much irrelevant information and invisible to small objects. In comparison, the too-small receptive fields will ignore the global relationships between pixels. Hence, capturing multiscale representations is desired for accurate and robust visual identification. Considering the scale-variability problem depicted above, we design a simple yet effective deformable attention module (DAM) to adjust its receptive field automatically. In addition, not all spatial information in

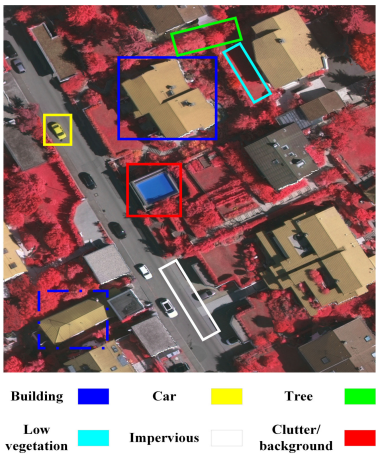


Fig. 1. Example of the complex spatial information and multiscale phenomenon in HRRS images.

HRRS images contributes equally to the semantic segmentation tasks. Hence, information from similar class-specific regions will be interfered with by irrelevant details. For example, for the areas containing the building category, only the surrounding areas of buildings are helpful for classification, and the rest areas may be redundant regions. This sparse class-specific phenomenon is similar to neuroscience. When the visual information is transmitted to the human brain, the visual cortex focuses on a special local area for a reference pixel.

Intuitively, the semantic segmentation tasks remain challenging to recognize contextual semantic information due to various geometric variations, e.g., object scale, pose, viewpoint, and partial deformation. Different from transformation-invariant features [19] or desired variations with sufficient data, Deformable ConvNet (DCNv1) [20] learns an input-specific 2-D spatial offset to model geometric transformation. The modulated Deformable ConvNet v2 (DCNv2) [21] contains learned channelwise attention to alleviate the influence of the irrelevant content by multiplying a quite small factor.

As illustrated in Fig. 2, the deformable convolution operation is more sensitive to various conditions, such as objects' scales and rotations.

In current visual recognition tasks, DCNNs extract multiscale features of the target through a layer-by-layer abstraction, which helps increase the richness and diversity. The shallower feature map can capture detailed information, such as boundaries and locations, but have lower semantics information. As the depth of the network increases, the more profound feature map loses some spatial information but contains more abstract, discriminative, and target-oriented semantic information, as shown in Fig. 3. Considering the semantic gap between low- and high-level features, we inject the proposed DAM at different stages of ResNet50 to achieve a balance.

For better simulating geometric transformation and removing the memory bottleneck, the spatial attention mechanism is reformulated by considering deformable convolutions operation with only memory cost $o(kn), k \leq n$. n generally refers to the sequence length. The proposed multiscale deformable attention net (MDANet) achieves excellent performance in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

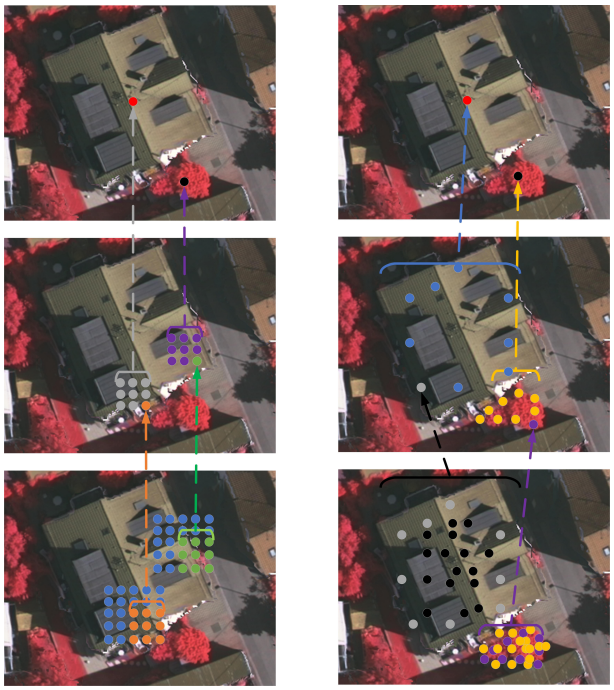


Fig. 2. Comparison between (Left) fixed receptive field in standard convolution and (Right) adaptive receptive field in deformable convolution.

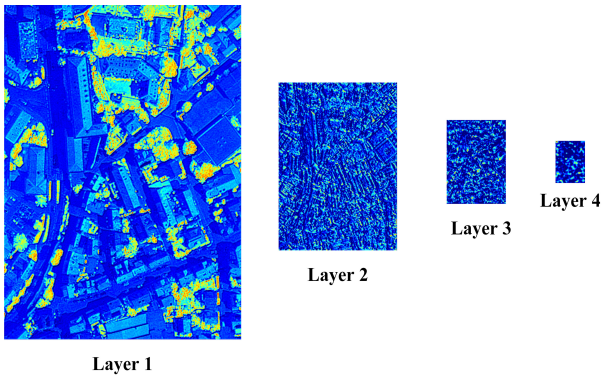


Fig. 3. Visualization of feature maps output by different layers in ResNet50.

mining spatial context information by focusing on the following aspects. First, we design a U-shaped architecture with skip connections to fuse high-level abstract features and low-level spatial features in HRRS images. Specifically, the DAM is embedded into different convolutional layers of the pretrained ResNet50 [22]. Besides, deformable convolution adaptively captures sampling pixels according to the complex spatial structures to enhance geometric extraction as much as possible. Finally, the extracted information will have abundant ground objects and complex spatial distribution if fusing multilayer features. The learnable spatial attention module is responsible for assigning independent priorities to any sampling pixels and adaptively learning each sampling pixel's contribution to predicting the reference pixel. Intuitively, the main idea is inspired by two common observations. First, the proposed DAM nestedly uses the idea that computes similarities between any pixels, which is of great significance for capturing the spatial relevances in HRRS images. From this perspective, our models are similar to previous attention-based methods, including CBAM, DANet, and LANet. Second, our work aligns with recent approaches that explored the use of "skip" connections between layers to capture multiscale features, including U-Net architecture. Compared with previous works, this article makes the following main contributions and innovations.

1) Different from the previous methods that induce attention weights over the whole input to prioritize relevant information for predicting, we rethink the use of spatial attention from a local perspective. Inspired by the deformable convolutions, for any reference pixel, the searches' space of the spatial attention module is restricted to a small set of its highly correlated pixels. To the best of our knowledge, the reformulated deformable convolution is first successfully introduced into the spatial attention module to remove the memory bottleneck and learn the geometry variability of objects.
2) In contrast to the deformable convolution, it is worth noting that the additional learned offsets offer spatial flexibilities to a reference pixel rather than a convolution kernel.
3) Unlike deformable DETR, the learned attention weight can be captured by the inner products instead of a separated convolution layer.
4) According to the common multiscale properties of HRRS images, our final contribution lies in direct inserting DAM into each stage of the ResNet50, aiming to capture sufficient spatial statistics and contextual cues with different scales. Meanwhile, the unified DAM can be trained directly with the backpropagation algorithm and conveniently embedded into any existing convolutional layer. Theoretically, it is easy to apply DAM to specific downstream tasks and applications.

The structure of this article is given as follows. Section I gives a brief introduction of the background and motivation of this article. Section II introduces the related works. Section III presents the details of the proposed DSM and MDANet. In Section IV, the experimental results and analysis are reported. Finally, the conclusion is made in Section V.

II. Related Works

A. Semantic Segmentation for HRRS Images

The semantic segmentation task aims to estimate a specific category to each pixel in an image, which is beneficial for analyzing and interpreting HRRS images. The emergence of the DCNNs has provided a new opportunity for mining and extracting efficient visual geographic features.

Parallel to the big achievements of DCNNs in computer vision (CV), geoscientists have discovered that DCNNs are suitable for processing HRRS images in which pixels are arranged regularly. Extracting features from HRRS images is, however, facing unprecedented and significant challenges.

The issues specific to DCNN-based HRRS image classification have been addressed in recent years.

1) Overcoming the difficulty of various complex geographic objects by the fusion of features from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

multiple resolutions. For example, the multiscale feature fusion (MFF) module [23] incorporates multiscale features generated by the high- and low-level features while restoring the image resolution. Peng *et al.* [24] further adopt multiple filters with different sizes to widen the network and increase the richness and diversity of the extracted features. Shang *et al.* [25] employ two atrous convolutions layers with different dilatation rates and global average pooling (GAP) to extract multiple scales' context information. However, the contextual dependence for the whole HRRS image are assumed homogeneous and nonadaptive.

2) Training new network instead of the fine-tuning professional network in CV domain [26].
3) Handling the problem of imbalanced distributions by designing a medium-frequency balancing strategy. For example, a neighboring pixel affinity loss (NPALoss) [27] is proposed to mitigate poor results for small-sized objects. A class attention module 28] is proposed to force the network to focus on small sample categories. Rao *et al.* [29] creatively modify the focal loss function to intensively center the mutual influence between unbalanced-class and complex land-cover diversities.
4) Alleviating the blurry object boundaries by strengthening boundary information and fine details. For example, the coordinate convolution [30] can significantly enhance the edge information of roads. Mi and Chen [31] mainly add boundary detection to the encoder–decoder architecture. An additional loss [32] is designed to guide the network that assigns a higher weight to the boundary pixels in ground-truth annotations, which retains object coherence and improves boundary accuracy.
5) Learning high-level features with limited labeled samples [33], [34]. For example, the combination of an improved self-taught feature learning framework and a semisupervised classification algorithm achieves the state-of-the-art performance on limited remote sensing datasets. Although the DCNNs have been widely used in HRRS image semantic segmentation, it still does not fully mind the spatial relationship cues between objects.

DL methods, such as deep belief networks (DBNs) [35], generative adversarial networks (GANs) [36], graph convolutional networks (GCNs) [37], and stacked autoencoders [38], also gained enormous achievements in semantic segmentation task for HRRS images.

B. Attention Mechanisms in Networks

For decades, the HVS has been extensively studied in cognitive psychology, neuroscience, philosophy, and computing. Recently, the visual attention mechanism inspired by HVS has become a predominant concept in neural network literature and becomes enormously popular for various applications, such as natural language processing (NLP) [39], CV [40], graph analysis [41], and multimodal tasks [42]. The pure self-attention vision model has also been verified to be an effective alternative to DCNNs [43], [44]. Besides, the visual attention mechanism provides greater insight into the interpretability of neural networks [45], [46].

We can understand the intuition behind the visual attention mechanism from the perspective of human biological systems. For example, the HVS is a cognitive process of tending to focus selectively on the target information while ignoring other irrelevant information. Similarly, the trainable visual attention mechanism has been successfully deployed to identify complex dependence between the elements (e.g., pixels and words) and selectively focus on the most task-relevant areas.

Attention mechanism was first introduced in NLP tasks. For example, Vaswani *et al.* [47] first present attention mechanisms to model alignment and retain history information for machine translation tasks. Exemplary performance from the attention mechanism on NLP has sparked great interest in CV tasks [48]. For example, a nonlocal operation [49] can capture the long-range spatial and channel dependence between any two signals. However, the attention mechanism has seldom been used in HRRS image pixelwise classification.

The ability of attention mechanism to learn dependence between elements and remarkable results has motivated outstanding HRRS image semantic segmentation efforts. For example, Xu *et al.* [50] design the lightweight dual attention module to obtain spatial relevance and dimension relevance. Xu *et al.* [51] utilize a self-attention mechanism to effectively establish relations between region features and relationships between local region features. Meanwhile, they also use cross-scale attention to select various spatial receptive fields adaptively. Niu *et al.* [52] present a class augmented attention (CAA) module to exploit the category-based correlation between pixels and recalibrate the class-level information. Ding *et al.* [53] employ patch-based attention to enhance the embedding of local contextual details.

C. Efficient Attention

For images composed of a relatively large number of pixels, the computing resource and quadratic time required by the vanilla dot-product attention mechanism are usually tremendous. Hence, several methods are developed to optimize the vanilla attention model.

Table I summarizes the recent attention model variants [54], classified by the technique characterization. Noted that n generally refers to the sequence length, and b is the local window size. For the efficient transformer models, the complexity mainly reflects the complexity of calculating attention matrix with specific parameterization. For example, the computation of vanilla dot-product attention requires that a $n \times n$ attention distribution matrix is stored. Thus, the computational complexity of vanilla dot-product attention is $o(n^2)$.

1) Fixed Patterns: The most straightforward modification restricts the search space for the attention mechanism to be fixed localized window or predefined block patterns. For example, a simple yet effective axial attention [55] applies an attention mechanism along a single axis of the tensor without flattening. Sparse transformer [56] separates the full self-attention operation across several steps. However, fixed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF EFFICIENT TRANSFORMER MODELS

Taxonomy	Models	Complexity
Low Rank/Kernels	Linformer	$o(n)$
	Synthesizer	$o(n^2)$
Fixed/Factorized/Random Patterns	Sparse Transformer	$o(n\sqrt{n})$
	Axial Transformer	$o(n\sqrt{n})$
Learnable Patterns	Reformer	$o(n\log n)$
	Routing Transformer	$o(n\log n)$
	Sinkhorn Transformer	$o(b^2)$
Memory	Set Transformer	$o(nk)$
	Memory Compressed	$o(nc^2)$
	Big Bird	$o(n)$

patterns can only apply attention mechanisms to fixed local regions without considering input sensitivity.

2) Learnable Patterns: End-to-end learnable patterns assign tokens to buckets/clusters in a data-dependent fashion, such as hash-based similarity measure [57] and online k-means clustering [58].

3) Low Rank: Another substantial research has been implemented based on low-rank assumptions. The intuition behind the method is that the matrix formed by the self-attention mechanism can be sufficiently approximated by multiple low-rank matrices, such as linear projection [59] and kernelization approximation [60]. However, those methods critically rely on low-rank assumptions.

4) Memory: End-to-end memory networks can be considered as a relatively new attention mechanism framework. It provides an explicit memory vector representation for each token and can compress the entire sequence. For example, inspired by sparse Gaussian process literature, set transformer [61] introduces an inducing point method for self-attention mechanism.

III. PROPOSED METHOD

A. Revisiting Self-Attention Block

The self-attention block is a critical component of transformer-based models. It mainly computes the relevance (weights) of any position to all positions and then summing all positions with corresponding weights. We briefly follow the generality to formulate the self-attention block.

For each input feature vector $x_i \in R^d$ in signal (e.g., image, sequence, and video), d indicates the feature dimension. The self-attention module first uses three trained linear embedding layers, W_θ, W_ψ, and W_φ, to convert x_i into three feature vectors: the query $q_i \in R^{d_k}$, the key $k_i \in R^{d_k}$, and the value $v_i \in R^{d_v}$

$$q_i = x_i W_\theta, k_i = x_i W_\psi, q_i = x_i W_\varphi\big). \tag{1}$$

Given the ith query element q_i and a set of key elements k_j, noted that subscript j is the index that enumerates all positions, and the set of compatibility scores between q_i and jth key k_j is measured by a pairwise dot product function $f(\cdot)$

$$f\big(q_i, k_j\big) = q_i^T \cdot k_j. \tag{2}$$

A SoftMax operation $g(\cdot)$ then normalizes the set of compatibility scores to obtain the attention weights

$$A_m(q_i, kj) = g\big(f\big(q_i, k_j\big)\big) = \frac{e^{q_i^T} \cdot k_j}{\sum_{\forall j} e^{q_i^T} \cdot k_j}. \tag{3}$$

The output signal y_q is generated as the sum of all the values v_i weighted by corresponding attention weights. Finally, the output signals in different heads are linearly aggregated with learnable weights W_m and W'_m.

A self-attention module for machine translation computes the responses at one position in a sequence by attending to all positions and weighted summing them up accordingly in an embedding space, which can be viewed as a form of nonlocal filtering operations that are applicable in CV. We follow the convention to formulate the self-attention module. Given an input signal (e.g., image, sequence, video, and feature) $X \in R^{n\times c}$, where n indicates the number of pixels in feature, and c is the number of channels, the output signal is generated as

$$y_q = \sum_{m=1}^{M} W_m \left[\sum_{k\in\Omega q} A_m\big(q_i, k_j\big) \odot W'_m v_i \right] \tag{4}$$

where m indexes the attention head and different attention heads learn various aspects in the model. Ωq represents the supporting search region for the query q_i

B. Enhanced Self-Attention for Localness Modeling

Although the self-attention mechanism shows its benefit in long-range interactions, it fully considers all the signals with a weighted sum operation. When processing HRRS images, its resource demands get prohibitively high. Meanwhile, we observe that most of the dependence relations occur in the adjacent pixels. Therefore, locality remains a solid inductive bias, and locality modeling is beneficial for semantic segmentation tasks. Mathematically, the number of dependence relations between pixels should be far lower than $o(\mathrm{HW})^2$. Several recent studies [53], [62] indicate that most attention layers learn to attend mainly to local neighborhoods since most data present a strong property of locality.

Compared with the vanilla attention mechanism that lacks any assumption about the data structuration, we make solid

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

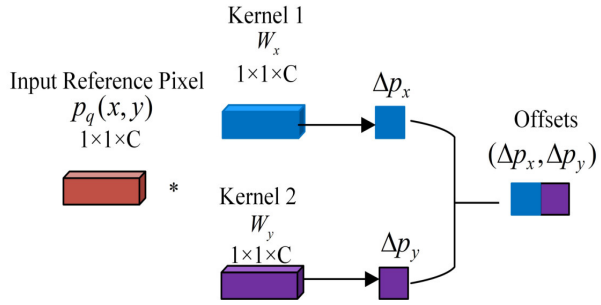


Fig. 4. Illustration of the proposed spatial sampling module. For the convenience of description, we assume that $K = 9$. Such a convolution layer can provide geometric transformation information and introduce locality into the network.

independent assumptions on the interactions between the attention variable.

Assume 1: Based on Tobler's first law of geography, we assume that each attention variable is independent in any other spatial context

$$p(A|X) = \prod_{i=1,j=1}^{i=m,j=n} p\big(a_{ij}|\delta(x_{i,j})\big). \tag{5}$$

The self-attention mechanism lacks generalization ability due to its unsuitable inductive bias about structuration. In this article, we further develop the deformable attention to optimize the self-attention mechanism by introducing locality constraints.

C. Deformable Spatial Attention

It is difficult to directly extract effective features from the whole HRRS images due to complex spatial feature relationships. Inspired by ConvNet and Deformable DETR [63], a new module is proposed to enable the network to focus on a small set of highly correlated pixels around any reference pixel. The intuition behind the new module is similar to the principles of spatial autocorrelation, which assumes that closer pixels are more similar compared to farther ones.

This section mainly describes the DAM from the following three dimensions: spatial sampling, distribution function, and fine-grained fusion.

1) Spatial Sampling: Specifically, given a reference pixel p_q with location (x, y) and content feature $p_q(x, y) \in R^{1\times1\times C}$ in an input feature map $A \in R^{H\times W\times C}$, where C and $H \times W$ denote the channel and spatial dimensions, the input-specific offsets $(\triangle p_x, \triangle p_y)$ can be learned via a separate $1 \times 1 \times C$ convolution layer with two output channels applied over $p_q(x, y)$, as illustrated in Fig. 4.

The learned convolution kernels can be updated using traditional backpropagation algorithm to adapt to diverse visual patterns concerning spatial positions. Considering that, we reformulated the output of $1 \times 1 \times C$ convolution operation to explain the intuition behind 2-D spatial sampling as follows:

$$(\triangle P_x, \triangle P_x) = \big(p_q(x, y)w_x^k\big), p_q(x, y)w_y^k\big)_{k=1}^{K} \tag{6}$$

where k enumerates all the sampled pixels, K is the total sampling pixels ($K \le H \times W$), and $(\triangle P_x, \triangle P_x)$ is a collection of sampling offsets. The spatially adaptive $1 \times 1 \times C$ convolution weights (w_x^k, w_y^k) for generating the kth offset can enable the sampling locations to be transferred to neighboring pixels. For the sake of simplicity, the bias term is omitted.

Then, the learned offsets $(\triangle P_x, \triangle P_x)$ are used for obtaining the values of the new sampling pixels located at an irregular position

$$p_k(x_k, y_k) = p_q\big((x + \triangle p_x), \big(y + \triangle p_y\big)\big). \tag{7}$$

Since the offsets might be fractional, the value of $p_q((x + \triangle p_x), (y + \triangle p_y))$ remains to be determined. In this case, bilinear interpolation $G(\cdot)$ is applied to determine the missed fractional value by its several neighboring grid pixels. For clarity, the value of $p_k(x_k, y_k)$ can generally be approximate as

$$\begin{aligned} p_k(x_k, y_k) &= \sum_{(p_{\text{tx}}, p_{\text{ty}})\in\tau} G\big((p_{\text{tx}}, p_{\text{ty}}), (x_k, y_k)\big) p_q\big(p_{\text{tx}}, p_{\text{ty}}\big) \\ &= \sum_{(p_{\text{tx}}, p_{\text{ty}})\in\tau} \max(0, 1 - |p_{\text{tx}} - x_k|) \\ &\quad \times \max\big(0, 1 - |p_{\text{ty}} - y_k|\big) p_q\big(p_{\text{tx}}, p_{\text{ty}}\big) \end{aligned} \tag{8}$$

where (x_k, y_k) represents an arbitrary position, τ enumerates all integer spatial locations clique near p_q in the input feature map, and the linear interpolation kernel $G(\cdot, \cdot)$ is decomposed in the x- and y-directions. Then, k neighboring sampling pixels will be concatenated to generate the corresponding sampling feature map $S \in R^{K\times C}$.

The spatial sampling module can automatically accommodate geometric variations of objects and encode the important local structure for each pixel. However, separate convolutions $1 \times 1 \times C$ obtain the input-sensitive 2-D spatial offsets without considering the information interaction between adjacent pixels. To this end, we introduce an attention module to model the interaction between the reference pixel and corresponding sampling pixels.

2) Distribution Function: The pixels extracted by the 2-D spatial sampling module contain more information about various geometric transformations. Moreover, the spatial attention module is integrated to extract and express the dependence between the reference pixel and sampling pixels. We convert the sampling feature map into two feature vectors, the Keys (K) and the Values (V). For clarity, we omit the coordinates corresponding to each variable. For example, we denote $p_q(x, y)$ as p_q. The dot product between p_q and the transpose of $p_k \in K$ is taken to calculate the relevance of the specific reference pixel and all sampling pixels. The softmax operation is then chosen as the distribution function to constrain each relevance between 0 and 1. Thus, the attention weights $A_{\text{qk}} \in R^{1\times K}$ can be defined as follows:

$$A_{\text{qk}} = \frac{\exp\big(p_q p_k^T\big)}{\sum_{k=1}^{K} \exp\big(p_q p_k^T\big)} \tag{9}$$

where spatial weight matrix A_{qk} measures the specific reference pixel's impact on the kth sampling pixels. The attention weights are then used for feature recalibrations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

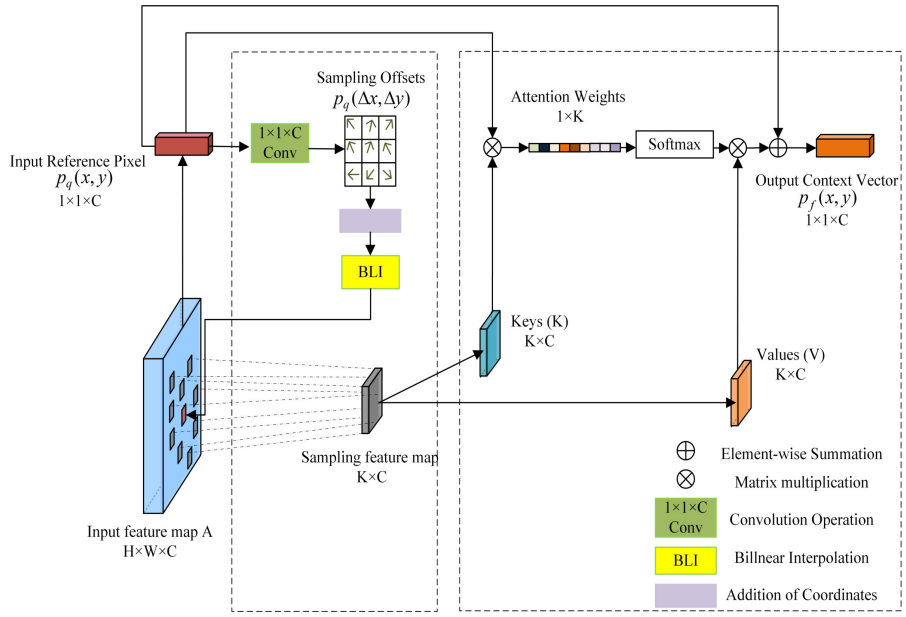


Fig. 5. Illustration of the proposed DAM. For the convenience of description, we assume that $K = 9$. We mainly insert the attention module after the output of the deformable convolution unit to refine the extracted features. The feature maps are shown as the shape of their tensors.

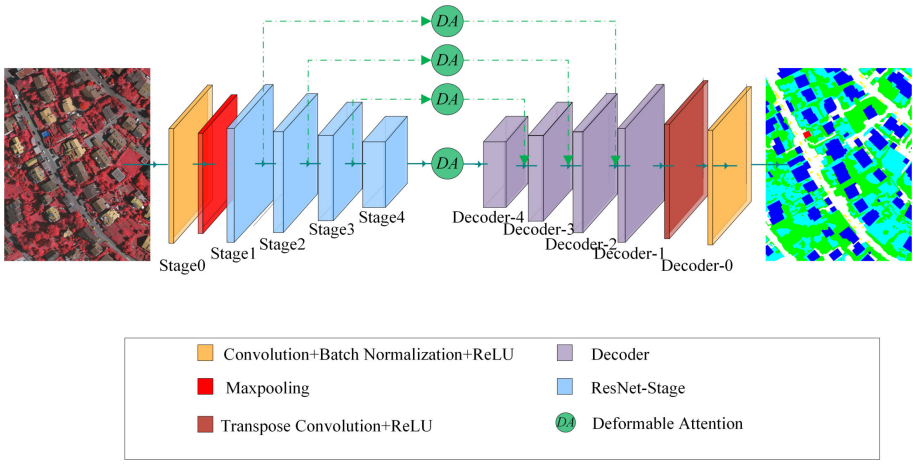


Fig. 6. Illustration of the proposed MDANet.

3) Fine-Grained Fusion: To selectively aggregate contexts and retain more semantic information in a global view, we extract context information by summing of all sampling pixels p_v with corresponding attention weights A_{qk}. Finally, the context information is combined with the original reference pixel to preserve certain initial features. The output through the fine-grained fusion module can be defined as follows:

$$p_f = \alpha \sum_{k=1}^{K} A_{\mathrm{qk}} p_v + p_q \tag{10}$$

where α is a scale parameter and increased gradually from 0 during training. Through the distribution function module and fine-grained fusion module, the sampling pixels and corresponding reference pixels can be better interacted with each other and have a more accurate feature representation. Since the dimensions of the input and output of the module are consistent, the DAM can be embedded in any network layer. The architectural view of the proposed DAM is illustrated in Fig. 5.

D. MDANet

Based on the proposed DAM, we design an MDANet. In MDANet, the DAM is embedded into different layers of ResNet50 to capture detailed location information and abstract semantic information, as shown in Fig. 6. The skip connection in different layers can extract the multiscale feature information to obtain detailed feature information and characterize objects at different scales.

IV. Experimental

A. Dataset

The proposed MDANet is evaluated on the two benchmark datasets: Vaihingen and DeepGlobe Land Cover Classification datasets.

1) Vaihingen Dataset: The Vaihingen dataset with a spatial resolution of 9 cm acquired by airborne sensors shows a relatively small village with many detached multistory buildings, more vegetation cover, and less traffic. As shown in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

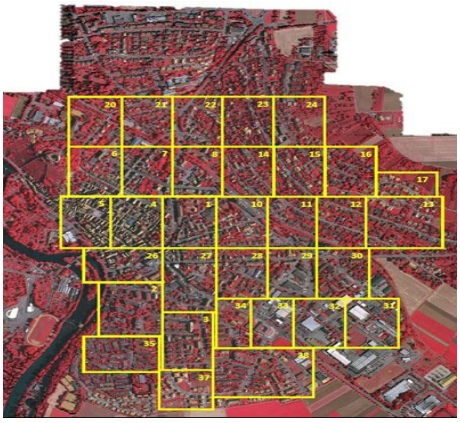


Fig. 7. Overview of the ISPRS Vaihingen 2-D dataset. The number in the upper right corner of each tile refers to the individual patch numbers as written at the filename endings.



Fig. 8. Overview of the land cover classification images is superimposed on corresponding satellite images, which covers 1717 km^2.

Fig. 7, there are 33 true orthophotos (TOPs) with an average size of 2494 × 2064 pixels, containing three channels: near-infrared (NIR), red (R), and green (G) bands. The corresponding digital surface model (DSM) derived from dense image matching techniques and the normalized DSM (NDSM) data are also provided. Please note that we do not use DSM and NDSM in our experiments for simplicity. Vaihingen datasets contain six object categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. Since the clutter/background category accounts for a tiny percentage of pixels, we ignore it during the quantitative evaluation. In addition, 25 TOP files are randomly used for training, and the remaining eight TOP tiles (ID: 1, 3, 5, 7, 11, 21, 15, and 30) are used to test the model.

2) DeepGlobe Land Cover Classification Dataset: The DeepGlobe Land Cover Classification dataset is the first public dataset offering high-resolution submeter satellite imagery focusing on rural areas, as shown in Fig. 8. Due to the variety of land cover types and the density of annotations, this dataset is more challenging than the Vaihingen datasets. DeepGlobe Land Cover Classification dataset contains 1146 visible-light satellite images of size 2448 × 2448 pixels in total with a pixel resolution of 50 cm; 803 of these images have publicly available human-annotated semantic segmentation labels covering seven land cover classes: urban, agriculture, rangeland, forest, water, barren, and unknown for nonland cover regions. For training, we randomly select 602 image-annotated images for training and the remaining 201 for testing. Please note that the "Unknown" class is not considered in the evaluation due to its tiny percentage of pixels.

B. Implementation Details

All the experiments are implemented on an Ubuntu 18.04 platform equipped with two Nvidia GeForce 1080Tis (11G). Meanwhile, all the operators are implemented with the PyTorch DL framework. For training, the adaptive moment estimation (Adam) is selected as an optimizer with an initial learning rate of 0.001 and a weight decay of 0.00001. The number of training epochs for the Vaihingen dataset is 50 for convergence, and the batch size is 12. The number of training epochs for the Deepglobe Land Cover Classification dataset is 80 due to more training samples. The dropout technique with a ratio of 0.5 is adopted to alleviate the overfitting phenomenon. Meanwhile, we use batch normalization to speed up the training process.

Training an entirely new model is infeasible due to limited data in most remote sensing studies. Meanwhile, to accelerate the training procedure and strengthen the robustness of the model, we use the initializing weight of a preexisting ResNet50 on the ImageNet dataset. Then, we fine-tune the model on the Vaihingen and Deepglobe datasets. A series of data augmentation transformations preprocess the training set. Specifically, we employ affine, scaling, flipping, and rotation to decrease the phenomenon of overfitting. Due to the limitation of computational resources, the input data are cropped using a sliding 864 × 864 window in a nonoverlapping fashion during the training phase.

C. Evaluation Metrics

To quantitatively verify the effectiveness and robustness of the MDANet model, we adopt the following evaluation metrics, including per-class IoU scores, mean intersection over union (mIoU), per-class F1 scores, mean F1 scores overall accuracy (OA), average classification accuracy (AA), average classification validity (AV), and kappa coefficient with the following formulas:

$$\mathrm{IoU}_h = \frac{p_{\mathrm{hs}}}{\sum_{s=1}^{N} p_{\mathrm{hs}} + \sum_{s=1}^{N} p_{\mathrm{sh}} - p_{\mathrm{hh}}} \tag{11}$$

$$\mathrm{mIoU} = \frac{1}{N}\sum_{h=1}^{N} \mathrm{IoU}_h \tag{12}$$

$$\mathrm{OA}_h = \frac{\sum_{h=1}^{N} p_{\mathrm{hh}}}{\sum_{h=1}^{N}\sum_{s=1}^{N} p_{\mathrm{hs}}} \tag{13}$$

$$\mathrm{AA}_h = \frac{1}{N}\sum_{h=1}^{N} \frac{p_{\mathrm{hh}}}{\sum_{s=1}^{N} p_{\mathrm{hs}}} \tag{14}$$

$$\mathrm{AV}_h = \frac{1}{N}\sum_{h=1}^{N} \frac{p_{\mathrm{hh}}}{\sum_{s=1}^{N} p_{\mathrm{sh}}} \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

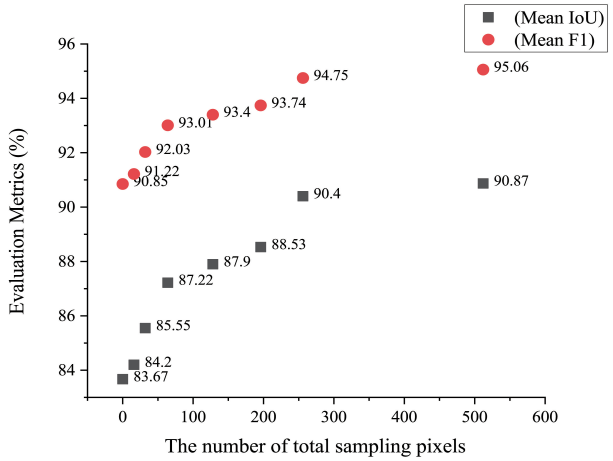


Fig. 9. Comparison between different K's in terms of MF1 and MIoU.

$$\text{kappa}_h = \frac{\sum_{h=1}^{N} p_{\text{hh}} \sum_{h=1}^{N} \sum_{s=1}^{N} p_{\text{hs}} - \sum_{h=1}^{N} p_{\text{hs}} p_{\text{sh}}}{\left(\sum_{h=1}^{N} \sum_{s=1}^{N} p_{\text{hs}}\right)2 - \sum_{h=1}^{N} p_{\text{hs}} p_{\text{sh}}} \tag{16}$$

$$\text{mF1} = \frac{1}{N} \sum_{h=1}^{N} \frac{2 \sum_{h=1}^{N} p_{\text{hh}}}{\sum_{s=1}^{N} p_{\text{hs}} + \sum_{s=1}^{N} p_{\text{sh}}} \tag{17}$$

where N is the number of categories of the pixels, and p_{hh}, p_{ss} is the number of pixels belong to hth, sth class being correctly classified to hth, sth class, respectively. p_{hs}, p_{sh} is the number of pixels belong to hth, sth class being wrongly classified to sth, hth class, respectively.

D. Results and Analysis

1) Ablation Study on Sensitivity Analysis of Critical Parameters: The most significant free hyperparameter in MDANet is the total sampling pixels K on each feature layer. From Section III-C, we can see that the predefined K determines the receptive field according to the objects' scale and shape. Hence, sensitivity analysis is performed to examine the influence of K on the model and choose the most suitable parameters. Specifically, we set K to 0, 16, 32, 64, 128, 196, 256, and 512 and keep other configurations consistent. Noted that, when K is set to 0, the model is equal to ResNet50.

As shown in Fig. 9, it is evident that MF1 and MIoU increase monotonically with K in the evaluation. However, when K increases to a certain degree, the rate of performance increase tends to decline may due to optimization challenges. Meanwhile, larger K will introduce more computational and memory complexities. Our results show that, when K is set to 512, the model can reach a relative balance between effective receptive fields and complexities.

2) Ablation Study on Individate Component: To stepwise quantitative validate the individual contribution of the proposed modules (DAM and MSS) to the final performance and the complementary effects between modules, ablation experiments are performed on the Vaihingen dataset. Table II presents the results for various design choices of the proposed MDANet. As reported in Table II, compared to the baseline (RseNet50), using an MSS, which allows information flow through different layer levels, can bring an additional 3.58% improvement in Miou scores and 2.09% improvement in MF1 scores. Noted that the DAM is embedded into the final layer of the pretrained ResNet50. As a comparison, after only introducing a DAM, the model yields an approximate increase of 7.77% in Miou scores and 4.52% in MF1 scores. It can adaptively reflect the complex structure more effectively and better explore the relevance of information between pixels. The DAM contributes to the results more than the MSS when combined. The proposed DMANet with both modules promotes a 0.19% in OA score and a 0.22% in F1 score, respectively, which has received the best performance compared to individual components.

TABLE II
ABLATION EXPERIMENTS FOR DAM ON VAIHINGEN DATASETS

Number	DAM	MSS	MIoU(%)	MF1(%)
1	×	×	83.67	90.85
2	×	✓	87.25	92.94
3	✓	×	89.30	94.17
4	✓	✓	91.44	95.37

TABLE III
ABLATION EXPERIMENTS FOR MSS ON VAIHINGEN DATASETS

MSS	MIoU(%)	MF1(%)	Memory(MB)
MSS-0	83.67	90.85	157.72
MSS-1	84.64	91.10	166.91
MSS-2	86.78	93.78	160.78
MSS-3	87.07	92.51	158.87
MSS-4	88.52	94.25	158.20
MSS-5	91.44	95.37	171.60

3) Ablation Study on MSS: In a multilayer neural architecture, different layers are projected into the pixel space and impact the output feature differently. Hence, we conduct the following ablation experiments on the Vaihingen dataset to verify the excellent performance brought by the proposed DAM at each stage of the ResNet50 and try to find the best appropriate stage combination. MSS-1, MSS-2, MSS-3, and MSS-4 refer to stage1, stage2, stage3, and stage4 inserted with DAM, while MSS-0 and MSS-5 refer to all stages inserted without, with DAM, respectively. Note that we do not consider the effects of stage0 due to its simple structure. From Table III, we can observe that the performances are highly related to the inserted stage for DAM. Specifically, it can be seen that the five MSS combinations improve the MIoU/MF1 by 7.77%/4.52%, 6.8%/4.27%, 4.66%/1.59%, 4.37%/2.86%, and 2.92%/1.22%, respectively, compared with the ResNet50. Our MDANet with all stages inserted by DAM outperforms any stage inserted by DAM. The difference may be because DAM is sensitive to the input feature resolution and a deeper level of neural architecture benefits more from the DAM in terms of quantitative quality. Meanwhile, the required computational memory for any combination of DAM on the 224 × 224 feature map is also shown. As a result, the DAMs are inserted in all stages in the following experiments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

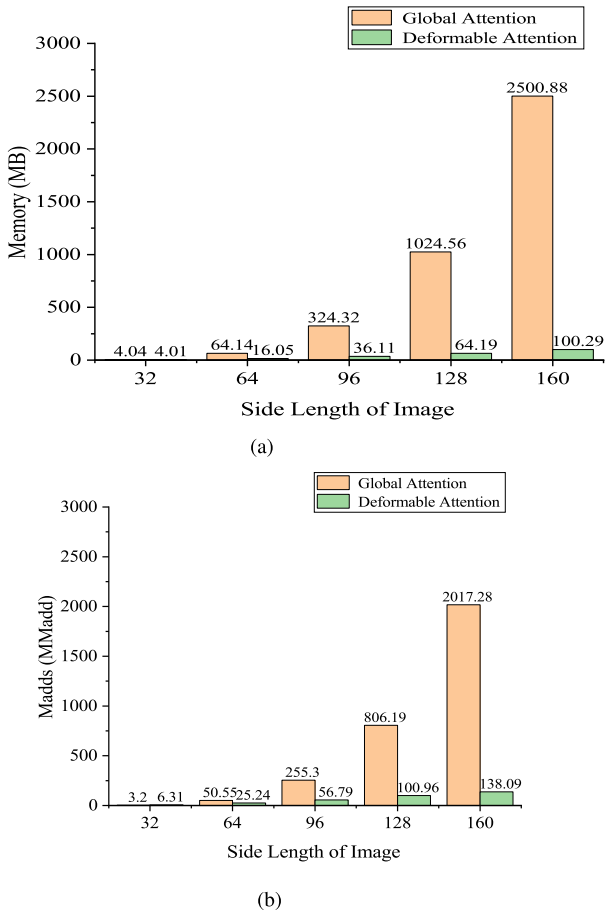


Fig. 10. Resource requirements under different input sizes. The green and orange bars depict the resource requirements of DAM and the global attention module, respectively. (a) Memory requirements. (b) Madds requirements.

4) Model Complexity: Considering that the complexity is significant to assess a model, we demonstrate the efficiency advantage of DAM over global attention module in memory with different sizes in Fig. 10. The basic parameters are abbreviated as memory, the unit is MByte (M), and the calculation amount is expressed by multiply–accumulate operations (Madds). For example, directly substituting the global attention module on the 64×64 feature map yields a four-time saving of memory and a two-time saving of Madds. The gap widens rapidly with the increase in the input size.

5) Performance Comparison: We perform abundant experiments on the state-of-the-art models to evaluate the performance of the proposed MDANet. The detailed quantitative comparison results are shown in Table IV. As depicted in this table, the MDANet achieves the most superior accuracy concerning mean IoU of 91.44%, mean F1-score of 95.37%, OA of 97.28%, AA of 95.50%, AV of 95.25%, and kappa of 95.97%. Comparisons with FCN, CBAM, PSPNet, DANet, ResNet50 (the backbone network), U-Net, LANet, where MDANet obtains increments of 17.51%, 15.10%, 13.54%, 4.88%, 7.77%, 12.69%, and 4.79% in mean IoU score, respectively. It is demonstrated that the MDANet can learn more discriminative feature representations by the fusion of multiscale features, locality modeling, and spatial relation modeling. Furthermore, the proposed model obtains the highest IoU scores and the highest F1 scores in most categories, especially for classes with good structural consistency and clear edges, such as buildings and cars. This verifies the spatial relationship information introduced by DAM enhanced the robustness in distinguishing various ground objects.

To further prove the generalization of MDANet, we conduct experiments on the Deepglobe Land Cover Classification dataset. The detailed quantitative results are shown in Table V. However, compared with FCN, CBAM, PSPNet, DANet, ResNet50, U-Net, LANet, and our proposed MDANet, we observed varying degrees of a performance drop (approximately −4.98%, −3.24%, −2.05%, −4.66%, −3.8%, −5.95%, −3.26%, and −3.12%) on the DeepGlobe dataset when comparing results on the Vaihingen 2-D dataset. The phenomenon is supported by the observation that there are some annotation inaccuracies, mainly introduced by highly ambiguous objects and lower ground resolutions. This obviously affected the model's final performance on the test sets. The proposed MDANet also achieves the highest Miou of 88.32%, MeanF1 of 93.60%, OA of 93.58%, AA of 94.64%, AV of 92.87%, and kappa of 91.87%. Compared with FCN, CBAM, PSPNet, DANet, ResNet50, U-Net, and LANet, the Miou has increased by 19.37%, 15.22%, 12.47%, 6.42%, 8.45%, 15.52%, and 4.93%, respectively. Compared with the baseline ResNet50, it can be seen that MDANet improved the Miou/MF1 of Urban, Agriculture, Rangeland, Forest, Water, Barren, and Unknown by 13.26%/8.05%, 9.54%/5.55%, 13.53%/9.92%, 5.63%/3.26%, 3.9%/2.07%, and 7.12%/3.89%, respectively, particularly a significant improvement in the Rangeland area. These results show that our proposed MDANet can better distinguish objects with variable target sizes.

An example of the visual comparisons of the various networks is emphasized in Fig. 11. We can see that the ground objects provided by FCN are mottled due to the loss of spatial information in the decoding process. For example, there is significant confusion between roads and low vegetation. In comparison, the results are more comprehensive after adding CBAM. The CBAM can obtain more effective features using average/max pooling to model the relations and dependence in spatial and channel dimensions. However, there are also lump-shaped misclassifications inside the building areas and low vegetation areas. The objects extracted by PSPNet are not very mottled, but the building has a jagged and blurring edge. Although PSPNet employs a pyramid pooling module to incorporate global image features and exploit multiscale local features, it ignores the detailed features due to GAP operations. It is observed that ResNet50 obtains more accurate and coherent segmentation results attributing to the deeper layers. However, due to the fixed sampling locations and the limited receptive field, the performance is unsatisfactory. For example, some pixels inside or on the edges of the buildings are misclassified as roads. With the fusion of high-level abstract features and low-level spatial features, U-Net can eliminate most noise and restore the boundary of the ground objects to a certain degree. However, a simple multilayer network cannot effectively utilize the abundant spatial and information in HRRS images. Observing from a larger scale, DANet shows

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

QUANTITATIVE METRIC COMPARISONS OF DIFFERENT METHODS ON THE VAIHINGEN DATASET

Method	Per-Class IoU Score(%)/Per-Class F1 Score(%)					Miou(%)	MeanF1(%)	OA(%)	AA(%)	AV(%)	kappa(%)
	Imp.surf.	Building	Low veg	Tree	Car						
FCN [64]	80.20/89.01	89.61/94.52	66.02/79.54	77.91/87.59	55.91/71.72	73.93	84.48	87.83	85.85	83.45	83.89
CBAM [65]	79.65/88.67	85.86/92.39	66.07/79.57	79.28/88.44	70.85/82.94	76.34	86.40	87.63	86.21	86.92	83.61
PSPNet [16]	88.44/93.87	91.10/95.34	77.05/87.04	83.42/90.96	49.52/66.24	77.90	86.69	91.62	84.49	90.48	88.90
DANet [66]	90.48/95.00	92.45/96.08	87.37/93.26	89.21/94.30	73.28/84.58	86.56	92.64	94.56	93.64	91.83	92.80
ResNet50 [22]	91.01/95.29	95.30/97.59	79.59/88.63	82.58/90.46	69.91/82.29	83.67	90.85	93.11	91.24	90.77	90.89
U-Net [67]	84.87/91.82	89.91/94.69	68.72/81.46	79.23/88.41	71.01/83.05	78.75	87.89	89.49	87.69	88.34	86.08
LANet [53]	89.30/94.35	92.37/96.03	85.83/92.37	89.05/94.21	76.73/86.83	86.65	92.76	94.21	92.71	92.34	92.34
MDANet	**95.70/97.80**	**97.67/98.82**	**92.68/96.20**	**93.51/96.64**	**77.61/87.40**	**91.44**	**95.37**	**97.28**	**95.50**	**95.25**	**96.40**

TABLE V

QUANTITATIVE METRIC COMPARISONS OF DIFFERENT METHODS ON THE DEEPGLOBE LAND COVER DATASET

Method	Per-Class IoU Score(%)/Per-Class F1 Score(%)						Miou(%)	MeanF1(%)	OA(%)	AA(%)	AV(%)	kappa(%)
	Urban	Agriculture	Rangeland	Forest	Water	Barren						
FCN	73.71/84.87	76.87/86.92	48.70/65.50	72.80/84.26	81.34/89.71	60.30/75.23	68.95	81.08	81.20	80.98	82.48	76.63
CBAM	70.40/82.63	81.79/89.98	51.48/67.97	79.42/88.53	89.55/94.49	65.95/79.48	73.10	83.85	85.09	83.67	84.70	81.27
PSPNet	75.88/86.28	81.49/89.80	47.67/64.56	82.17/90.21	90.17/94.83	77.77/87.50	75.85	85.53	86.60	85.41	86.30	83.14
DANet	80.12/88.96	86.95/93.02	60.28/75.22	83.69/91.12	95.00/97.43	92.38/96.04	81.90	89.66	89.92	90.11	89.65	87.27
ResNet50	74.95/85.69	80.62/89.27	58.49/73.81	83.10/90.77	94.32/97.08	87.72/93.46	79.87	88.35	88.89	88.62	89.38	86.05
U-Net	77.16/87.11	77.79/87.51	45.00/62.07	77.19/87.13	89.84/94.65	69.20/81.80	72.80	83.40	84.54	83.40	85.19	80.62
LANet	82.22/90.24	85.07/91.94	64.67/78.55	85.04/91.91	95.08/97.48	88.23/93.75	83.39	90.64	90.83	91.63	89.97	88.39
MDANet	**88.21/93.74**	**90.16//94.82**	**72.02/83.73**	**88.73/94.03**	**95.98/97.95**	**94.84/97.35**	**88.32**	**93.60**	**93.58**	**94.64**	**92.87**	**91.87**

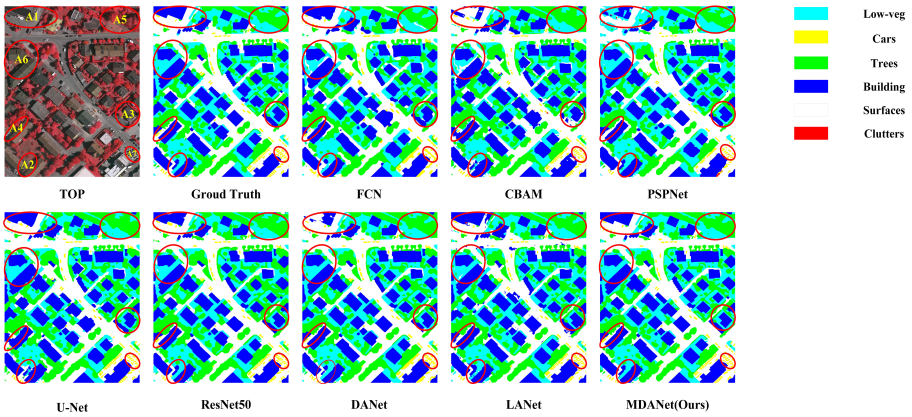


Fig. 11. Qualitative comparison of semantic segmentation results on the Vaihingen dataset (ID: 30) for various networks. We leverage the red ellipses to highlight the challenging regions that are easily misclassified (zoomed-in view to see more details).

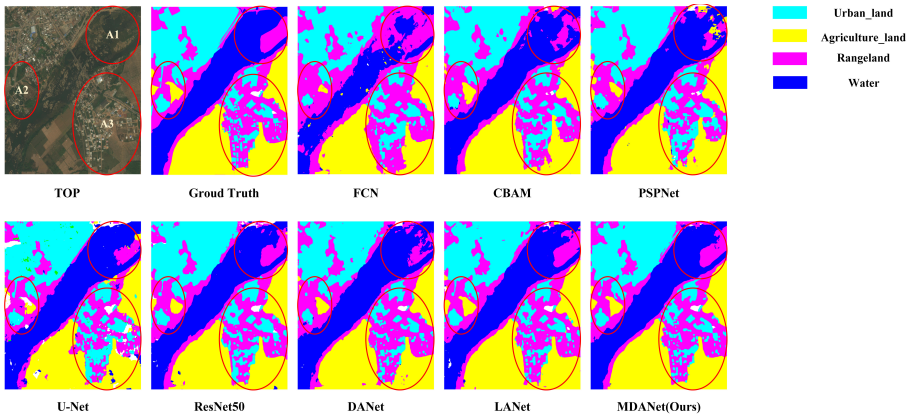


Fig. 12. Qualitative comparison of semantic segmentation results on the DeepGlobe Land Cover Classification dataset (ID: 799523) for various networks. The regions marked as A1 were covered with the confusing area, such as water and rangland. The regions marked as A2 and A3 were mostly covered with challenging ground objects, agriculture and urban, respectively. We leverage the red ellipses to highlight the challenging regions that are easily misclassified (zoom in to see more details).

improvements in segmenting confusing areas due to long-range modeling but produces fragmented segments. With the aggregation of local contextual information in different layers, LANet significantly reduces the errors and better preserves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

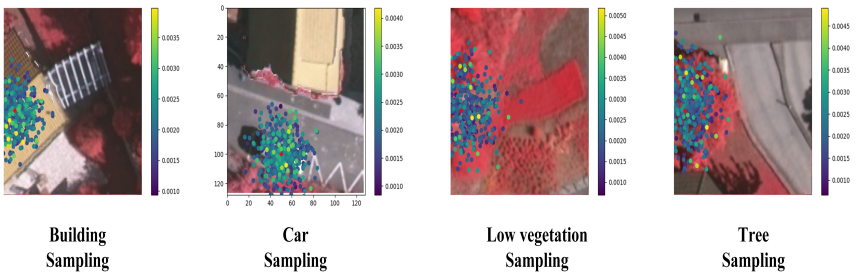


Fig. 13. Visualization of effective sampling locations in reformulated DAM. Each image shows the sampling locations (512 points) in the final layer of ResNet50 for four activation units (red cross marker) on the building, car, low vegetation, and tree, respectively. The modulated DAM enables models to focus on more class-specific details. Each sampling point is marked as a filled circle whose color indicates its corresponding attention weights.

the spatial details. However, there are several sporadic and miscellaneous classification errors inside the vegetation and building area.

In comparison, the proposed MDANet significantly preserves more delicate boundary information and maintains object integrity. The interactions between surrounding beneficial pixels can help facilitate the suppression of information that decreases the classification accuracy. For example, there are fewer false alarms in the surrounding areas of buildings. Meanwhile, it is more proper regard to the miniature objects. In addition, although trees and low vegetation are very similar in appearance and always appear in adjacent locations, MDANet still dramatically discriminates against them.

To better demonstrate the effectiveness of our MDANet, we visualize the qualitative comparison results between MDANet and other models in Fig. 12. We can see that the inconsistent results generated by FCN and U-Net have some serious missing semantic objects due to the loss of spatial information, particularly at the contour of objects. The CBAM reduces the incomplete semantic objects but produces boundary confusion due to the limited feature extraction ability. Although receptive-field-enlarging networks, such as PSPNet, demonstrate the ability to enhanced segmentation in confusing areas, the complex contour is not entirely generated. In addition, ResNet50 produces fewer fragments but fails to maintain consistency in larger areas, particularly mixing up similarly looking classes, such as water and rangland. The utilization of attention mechanisms, i.e., DANet, and LANet, brings a limited accuracy increase. DANet has a better detection effect on contour and detail areas over a large area, while the prediction map of the LANet has fewer broken segments and is more coherent. Beyond the models mentioned above, with DAM extracting spatial dependence and simulating geometric transformation in multilayers, our proposed MDANet relatively better preserves the geometric details and complex contours in both larger and detailed areas. Specifically, there is a significant improvement in preserving the boundaries and reducing fragmented segments of rangland. This verifies that our proposed MDANet enhanced the robustness in distinguishing various ground objects.

6) Visualization of Spatial Support in Multiscale DAM: To facilitate the understanding of the adaptive DAM more vividly in the last layer, four samples of deformable attention maps from different scales of objects (Car, Building, Low vegetation, and Tree) are displayed in Fig. 13. We chose 512 points, mainly because the model can reach a relative balance between effective receptive fields and complexities, as described in Section IV-D1. We can see that the network learns to concentrate on the most representative region for the specific class. The visualization also demonstrates that the reformulated DAM can adaptively capture geometric transformation and interaction between adjacent pixels according to the actual structure of ground objects.

V. Conclusion

In this article, we have proposed a novel adaptation according to the characteristics of HRRS images; this article mainly brings locality to the spatial attention module to alleviate the memory bottleneck and model complex geometric transformation. The proposed reformulated DAM is responsible for attending to any pixel's neighbor pixels and modeling interactions between them, which is beneficial for popular recognition tasks (e.g., image classification, object detection, action recognition, video processing, and segmentation). Benefiting from and DAM and the aggregation of low- and high-level features, the proposed MDANet offers competitive performance in our extensive experiments.

Meanwhile, several future research directions are in need to be investigated. An important consideration is to explicitly force a better alignment between HVS and visual attention mechanism and explore the interpretability of visual attention mechanism. Another future direction is to research the potential of the visual attention mechanisms for alternating DCNNs.

References

[1] L. Zhu, J. Suomalainen, J. Liu, J. Hyyppä, H. Kaartinen, and H. Haggren, "A review: Remote sensing sensors," in *Multi-Purposeful Application of Geospatial Data*. London, U.K.: InTech, 2018, pp. 19–42.

[2] L. Luo *et al.*, "Airborne and spaceborne remote sensing for archaeological and cultural heritage applications: A review of the century (1907–2017)," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111280.

[3] Q. Yuan *et al.*, "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[4] M. J. McCarthy *et al.*, "Satellite remote sensing for coastal management: A review of successful applications," *Environ. Manage.*, vol. 60, no. 2, pp. 323–339, 2017.

[5] X. Li, W. Y. Chen, G. Sanesi, and R. Lafortezza, "Remote sensing in urban forestry: Recent applications and future directions," *Remote Sens.*, vol. 11, no. 10, p. 1144, May 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

[6] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

[7] Y. Li, J. Ma, and Y. Zhang, "Image retrieval from remote sensing big data: A survey," *Inf. Fusion*, vol. 67, pp. 94–115, Mar. 2021.

[8] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, no. 99, pp. 3735–3756, Jun. 2020.

[9] J. Schmidhuber, "Deep learning in neural networks: An overview," *Neural Netw.*, vol. 61, pp. 85–117, Jan. 2015.

[10] J. Song, S. Gao, Y. Zhu, and C. Ma, "A survey of remote sensing image classification based on CNNs," *Big Earth Data*, vol. 3, no. 3, pp. 232–254, Jul. 2019.

[11] X. Lu, H. Sun, and X. Zheng, "A feature aggregation convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7894–7906, Oct. 2019.

[12] H. Sun, S. Li, X. Zheng, and X. Lu, "Remote sensing scene classification by gated bidirectional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 82–96, Jan. 2020.

[13] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[14] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4353–4361.

[15] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[16] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[17] J. Fu, J. Liu, Y. Wang, J. Zhou, C. Wang, and H. Lu, "Stacked deconvolutional network for semantic segmentation," *IEEE Trans. Image Process.*, early access, Jan. 25, 2019, doi: 10.1109/TIP.2019.2895460.

[18] G. W. Lindsay, "Attention in psychology, neuroscience, and machine learning," *Frontiers Comput. Neurosci.*, vol. 14, p. 29, Apr. 2020.

[19] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[20] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[21] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets V2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[23] X. He *et al.*, "Semantic segmentation of remote-sensing images based on multiscale feature fusion and attention refinement," *IEEE Geosci. Remote Sens. Lett.*, early access, Jan. 26, 2021, doi: 10.1109/LGRS.2021.3052557.

[24] C. Peng, Y. Li, L. Jiao, Y. Chen, and R. Shang, "Densely based multi-scale and multi-modal fully convolutional networks for high-resolution remote-sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 2612–2626, Aug. 2019.

[25] R. Shang, J. Zhang, L. Jiao, Y. Li, N. Marturi, and R. Stolkin, "Multi-scale adaptive feature fusion network for semantic segmentation in remote sensing images," *Remote Sens.*, vol. 12, no. 5, p. 872, Mar. 2020.

[26] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[27] Y. Feng *et al.*, "NPALOSS: Neighboring pixel affinity loss for semantic segmentation in high-resolution aerial imagery," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 5, no. 2, pp. 1–8, 2020.

[28] X. Li *et al.*, "Dual attention deep fusion semantic segmentation networks of large-scale satellite remote-sensing images," *Int. J. Remote Sens.*, vol. 42, no. 9, pp. 3583–3610, May 2021.

[29] Z. Rao, M. He, and Y. Dai, "Class attention network for semantic segmentation of remote sensing images," in *Proc. Asia–Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, 2020, pp. 150–155.

[30] S. Wang, H. Yang, Q. Wu, Z. Zheng, Y. Wu, and J. Li, "An improved method for road extraction from high-resolution remote-sensing images that enhances boundary information," *Sensors*, vol. 20, no. 7, p. 2064, Apr. 2020.

[31] L. Mi and Z. Chen, "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 140–152, Jan. 2020.

[32] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2018.

[33] K. Ronald, R. Luu, and C. Kanan, "Low-shot learning for the semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6214–6223, Oct. 2018.

[34] Z. Huang, Z. Pan, and B. Lei, "Transfer learning with deep convolutional neural network for SAR target classification with limited labeled data," *Remote Sens.*, vol. 9, no. 9, p. 907, 2017.

[35] C. Li, Y. Wang, X. Zhang, H. Gao, Y. Yang, and J. Wang, "Deep belief network for spectral–spatial classification of hyperspectral remote sensor data," *Sensors*, vol. 19, no. 1, p. 204, Jan. 2019.

[36] S. Xu, X. Mu, D. Chai, and X. Zhang, "Remote sensing image scene classification based on generative adversarial networks," *Remote Sens. Lett.*, vol. 9, no. 7, pp. 617–626, 2018.

[37] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 15, 2021, doi: 10.1109/TNNLS.2021.3071369.

[38] X. Dai, X. He, S. Guo, S. Liu, F. Ji, and H. Ruan, "Research on hyperspectral remote sensing image classification by applying stacked denoising auto-encoders neural network," *Multimedia Tools Appl.*, vol. 80, no. 14, pp. 21219–21239, 2021.

[39] A. Galassi, M. Lippi, and P. Torroni, "Attention in natural language processing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 10, pp. 4291–4308, Oct. 2021.

[40] X. Zhu, D. Cheng, Z. Zhang, S. Lin, and J. Dai, "An empirical study of spatial attention mechanisms in deep networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6688–6697.

[41] F. Ma, F. Gao, J. Sun, H. Zhou, and A. Hussain, "Attention graph convolution network for image segmentation in big SAR imagery data," *Remote Sens.*, vol. 11, no. 21, p. 2586, Nov. 2019.

[42] A. Zadeh, P. P. Liang, S. Poria, P. Vij, E. Cambria, and L.-P. Morency, "Multi-attention recurrent network for human communication comprehension," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 1–8.

[43] P. Ramachandran, N. Parmar, A. Vaswani, I. Bello, A. Levskaya, and J. Shlens, "Stand-alone self-attention in vision models," 2019, *arXiv:1906.05909*. [Online]. Available: http://arxiv.org/abs/1906.05909

[44] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*. [Online]. Available: http://arxiv.org/abs/2010.11929

[45] K. Mrini, F. Dernoncourt, Q. Tran, T. Bui, W. Chang, and N. Nakashole, "Rethinking self-attention: Towards interpretability in neural parsing," 2019, *arXiv:1911.03875*. [Online]. Available: http://arxiv.org/abs/1911.03875

[46] S. Wiegreffe and Y. Pinter, "Attention is not not explanation," 2019, *arXiv:1908.04626*. [Online]. Available: http://arxiv.org/abs/1908.04626

[47] A. Vaswani *et al.*, "Attention is all you need," 2017, *arXiv:1706.03762*. [Online]. Available: http://arxiv.org/abs/1706.03762

[48] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," 2021, *arXiv:2101.01169*. [Online]. Available: http://arxiv.org/abs/2101.01169

[49] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[50] Z. Xu, W. Zhang, T. Zhang, and J. Li, "HRCNet: High-resolution context extraction network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 13, no. 1, p. 71, Dec. 2020.

[51] Q. Xu, X. Yuan, C. Ouyang, and Y. Zeng, "Attention-based pyramid network for segmentation and classification of high-resolution and hyperspectral remote sensing images," *Remote Sens.*, vol. 12, no. 21, p. 3501, Oct. 2020.

[52] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 24, 2021, doi: 10.1109/TGRS.2021.3065112.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

[53] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.
[54] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," 2020, *arXiv:2009.06732*. [Online]. Available: http://arxiv.org/abs/2009.06732
[55] J. Ho, N. Kalchbrenner, D. Weissenborn, and T. Salimans, "Axial attention in multidimensional transformers," 2019, *arXiv:1912.12180*. [Online]. Available: http://arxiv.org/abs/1912.12180
[56] R. Child, S. Gray, A. Radford, and I. Sutskever, "Generating long sequences with sparse transformers," 2019, *arXiv:1904.10509*. [Online]. Available: http://arxiv.org/abs/1904.10509
[57] N. Kitaev, Ł. Kaiser, and A. Levskaya, "Reformer: The efficient transformer," 2020, *arXiv:2001.04451*. [Online]. Available: http://arxiv.org/abs/2001.04451
[58] A. Roy, M. Saffar, A. Vaswani, and D. Grangier, "Efficient content-based sparse attention with routing transformers," *Trans. Assoc. Comput. Linguistics*, vol. 9, pp. 53–68, Feb. 2021.
[59] S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma, "Linformer: Self-attention with linear complexity," 2020, *arXiv:2006.04768*. [Online]. Available: http://arxiv.org/abs/2006.04768
[60] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret, "Transformers are RNNs: Fast autoregressive transformers with linear attention," in *Proc. 37th Int. Conf. Mach. Learn.*, 2020, pp. 5156–5165
[61] J. Lee, Y. Lee, J. Kim, A. Kosiorek, S. Choi, and Y. W. Teh, "Set transformer: A framework for attention-based permutation-invariant neural networks," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 3744–3753.
[62] N. Parmar *et al.*, "Image transformer," in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, pp. 4055–4064.
[63] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*. [Online]. Available: http://arxiv.org/abs/2010.04159
[64] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[65] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
[66] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[67] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.



Renxiang Zuo received the B.Eng. degree in surveying and mapping engineering from the School of Geomatics Science and Technology, Nanjing Tech University, Nanjing, China, in 2019, where he is currently pursuing the master's degree in architectural and civil engineering.

His research interests include deep learning and remote sensing applications.



Guangyun Zhang received the B.Sc. degree from the China University of Mining and Technology, Xuzhou, China, in 2007, and the Ph.D. degree in electrical engineering from the University of New South Wales, Sydney, NSW, Australia, in 2013.

From 2014 to 2019, he was with the Remote Sensing Research Center, Tianjin University, Tianjin, China, where he is currently an Associate Professor. In 2019, he moved to Nanjing Tech University, Nanjing, China, where he is a Professor with the School of Geomatics Science and Technology. His research interests include imaging spectrometry and LiDAR.



Rongting Zhang received the B.S. degree in marine technology from the Tianjin University of Science and Technology, Tianjin, China, in 2012, the M.S. degree in geological resources and geological engineering from the Guilin University of Technology, Guilin, China, in 2015, and the Ph.D. degree in instrumentation science and technology from the School of Precision Instrument and Opto-Electronics Engineering, Tianjin University, Tianjin, in 2020.

He is currently a Lecturer with the College of Geomatics Science and Technology, Nanjing Tech University, Nanjing, China.



Xiuping Jia (Fellow, IEEE) received the B.Eng. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in 1982, and the Ph.D. degree in electrical engineering from the University of New South Wales, Canberra, ACT, Australia, in 1996.

Since 1988, she has been with the School of Information Technology and Electrical Engineering, University of New South Wales, where she is currently a Senior Lecturer. She is also a Guest Professor with Harbin Engineering University, Harbin, China, and an Adjunct Researcher with the National Engineering Research Center for Information Technology in Agriculture, Beijing, China. She is a coauthor of the remote sensing textbook titled *Remote Sensing Digital Image Analysis* (Springer-Verlag, Third Edition, 1999, and Fourth Edition, 2006). Her research interests include remote sensing, image processing, and spatial data analysis.

Dr. Jia is also a Subject Editor of the *Journal of Soils and Sediments* and also serves as an Associate Editor-in-Chief of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.